

Linear transformation

Basis vec. $\hat{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\hat{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

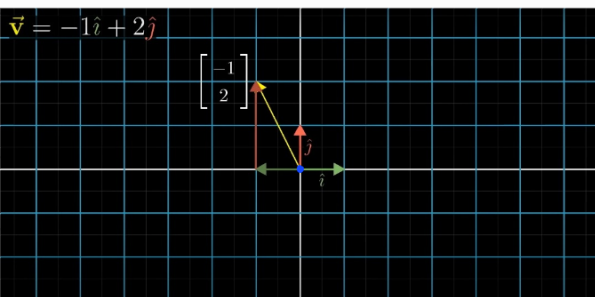
$$\underline{v} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} = (-1)\hat{i} + 2\hat{j}$$

New basis vec.

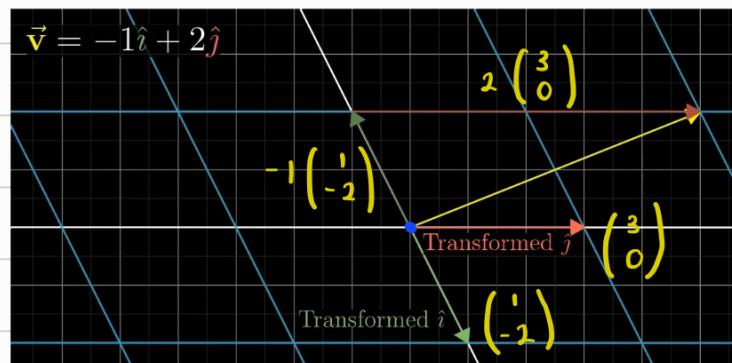
$$\underline{v}_{\text{transformed}} = (-1)\hat{i}_{\text{transformed}} + 2\hat{j}_{\text{transformed}}$$

So if after transformation, $\hat{i}_{\text{transformed}} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$, $\hat{j}_{\text{transformed}} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$

$$\Rightarrow \underline{v}_{\text{transformed}} = \left[-1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} + 2 \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right] = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$



Transformed



Generally, Let $\underline{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ undergo the same transformation above,

$$\Rightarrow \underline{v}_{\text{transformed}} = x \begin{pmatrix} 1 \\ -2 \end{pmatrix} + y \begin{pmatrix} 3 \\ 0 \end{pmatrix}$$

It is a common practice to package $\hat{i}_{\text{transformed}}$ and $\hat{j}_{\text{transformed}}$

into a 2×2 matrix

$$\Rightarrow \begin{pmatrix} \boxed{1} & \boxed{3} \\ \boxed{-2} & \boxed{0} \end{pmatrix}$$

$\hat{i}_t$ $\hat{j}_t$

So generally

Matrix vec. multiplication

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} a \\ c \end{pmatrix} + y \begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$$

OR

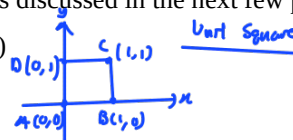
$$= \begin{pmatrix} (a, b) \cdot (x, y) \\ (c, d) \cdot (x, y) \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$$

$$= \begin{pmatrix} \text{row}_1 \cdot \text{column}_1 \\ \text{row}_2 \cdot \text{column}_2 \end{pmatrix}$$

Some Common Types of Transformation

We will be using the following examples for all transformations discussed in the next few pages:

- the unit square, with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$
- the triangle with vertices $(1, 1)$, $(3, 1)$, $(3, 2)$

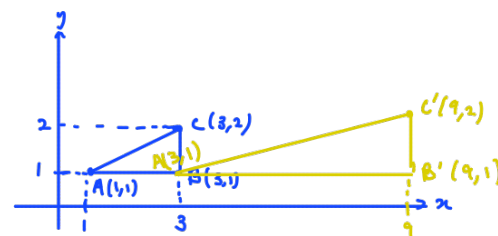
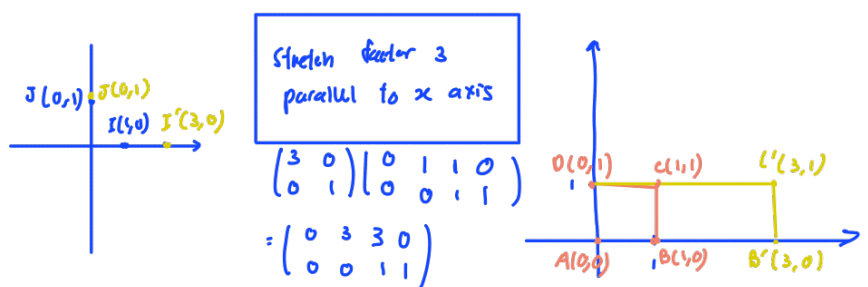


(I) Stretch (Dilation)

- Stretch by a scale factor of k parallel to the x-axis:

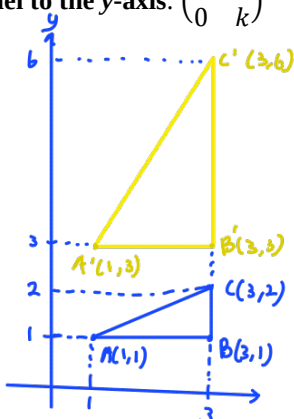
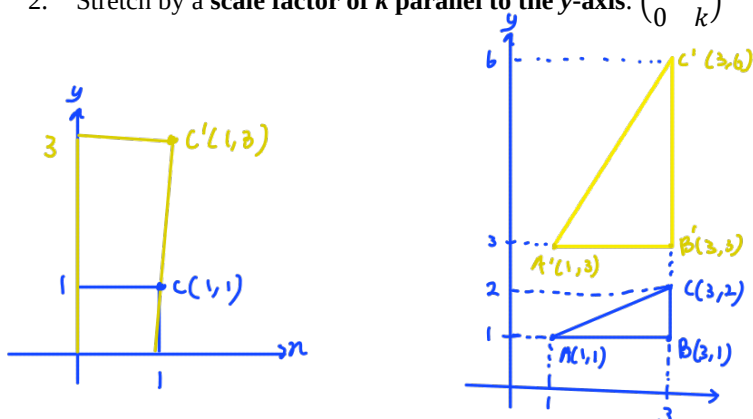
$$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$$

Describe the trans. represented by this matrix



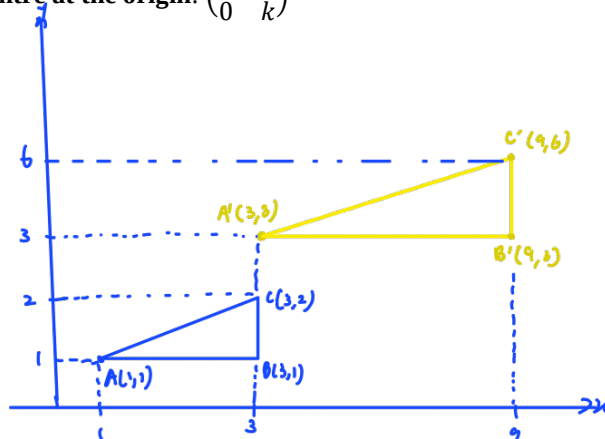
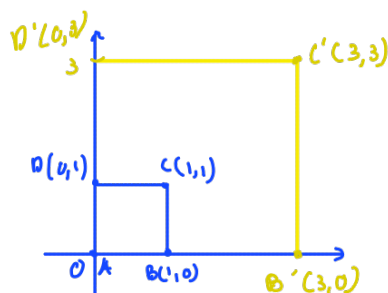
- Stretch by a scale factor of k parallel to the y-axis:

$$\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$$



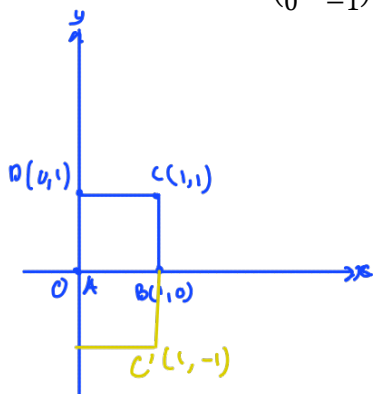
(II) Enlargement

Enlargement by a scale factor of k with centre at the origin: $\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$



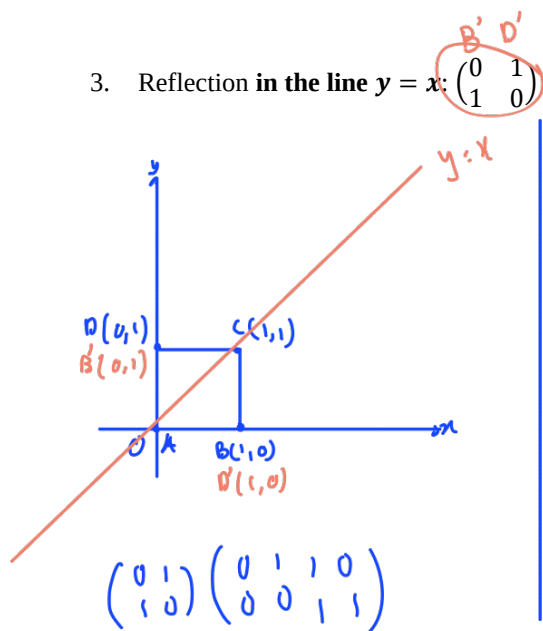
(III) Reflection

1. Reflection **in the x-axis**: $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

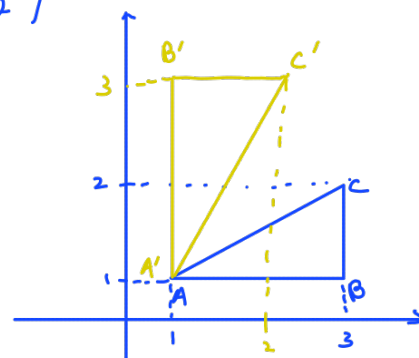


2. Reflection **in the y-axis**: $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

3. Reflection **in the line $y = x$** : $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$



$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 3 & 3 \\ 1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 3 & 3 \end{pmatrix}$$

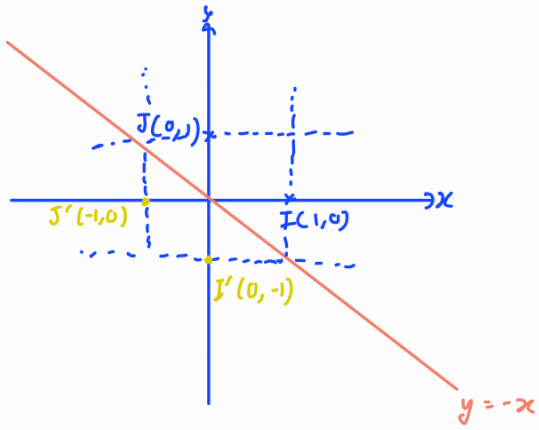


Note: The matrix which represents a reflection in the line $y = mx$ is given by $\frac{1}{1+m^2} \begin{pmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{pmatrix}$

$$\begin{aligned} y &= -3x \\ \text{Trans matrix} &= \frac{1}{1+(-3)^2} \begin{pmatrix} 1-(-3)^2 & 2(-3) \\ 2(-3) & (-3)^2-1 \end{pmatrix} \\ &= \frac{1}{10} \begin{pmatrix} -8 & -6 \\ -6 & 8 \end{pmatrix} \end{aligned}$$

Reflection in the line $y = -x$.

Find the transformation matrix

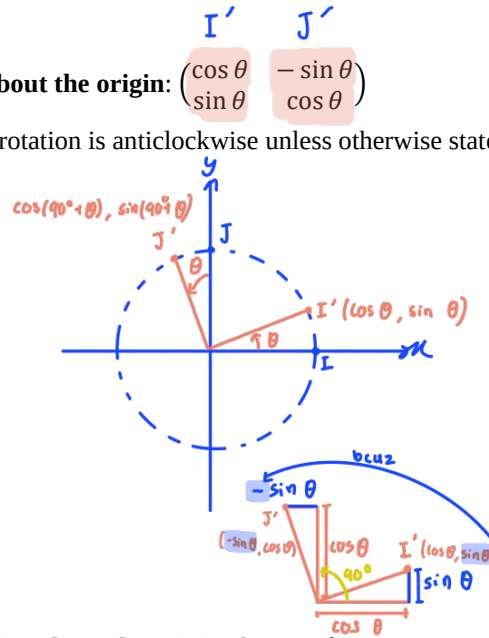
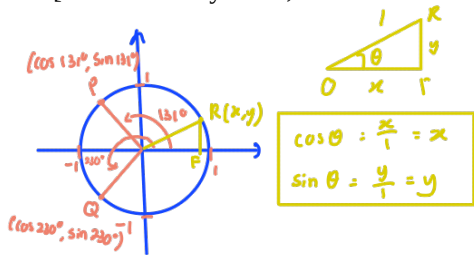


$$\text{Trans Matrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

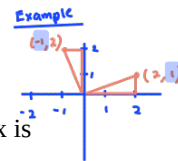
(IV) Rotation

Rotation through an angle θ **anticlockwise about the origin**: $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

[Note: In our syllabus, we will assume that a rotation is anticlockwise unless otherwise stated.]



$$\begin{aligned} J' &= [\cos(90^\circ + \theta), \sin(90^\circ + \theta)] \\ \cos(90^\circ + \theta) &= \cos 90^\circ \cos \theta - \sin 90^\circ \sin \theta \\ &= 0 \cos \theta - 1 \sin \theta \\ &= -\sin \theta \\ \sin(90^\circ + \theta) &= \sin 90^\circ \cos \theta + \cos 90^\circ \sin \theta \\ &= 1 \cos \theta + 0 \sin \theta \\ &= \cos \theta \end{aligned}$$

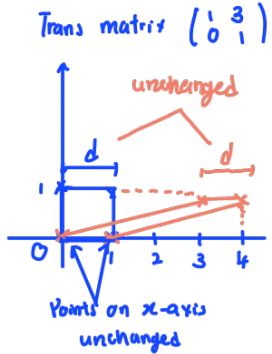


Note: For rotation through an angle θ **clockwise about the origin**, the transformation matrix is

$$\begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

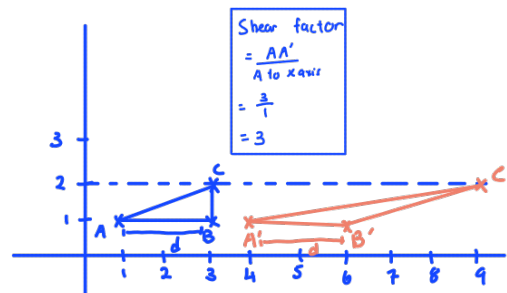
(V) Shear

1. Shear with shear factor k , **parallel to the x-axis**: $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$



$$\begin{aligned} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= \begin{pmatrix} 4 \\ 1 \end{pmatrix} \\ \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \begin{pmatrix} 3 \\ 1 \end{pmatrix} \\ \det F &= 1 \times 1 - 3 \times 0 \\ &= 1 \end{aligned}$$

Shear factor
= $\frac{\text{Dist bwn point \& image}}{\text{Dist of point from x axis}}$



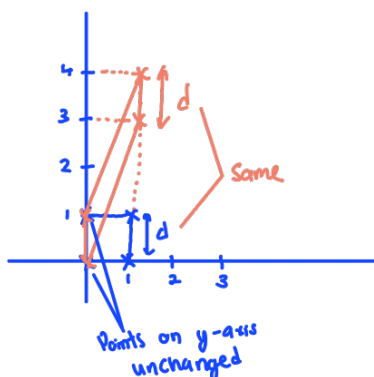
Shear factor
= $\frac{AA'}{A \text{ to } x\text{-axis}}$
= $\frac{3}{1}$
= 3

$$\begin{aligned} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= \begin{pmatrix} 4 \\ 1 \end{pmatrix} \\ \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} &= \begin{pmatrix} 6 \\ 1 \end{pmatrix} \\ \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} &= \begin{pmatrix} 9 \\ 2 \end{pmatrix} \end{aligned}$$

2. Shear with shear factor k , **parallel to the y-axis**: $\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$

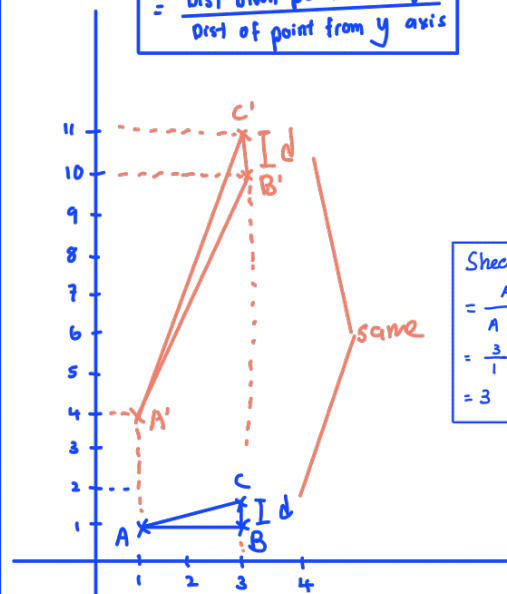
Trans matrix = $\begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$

Vertical



$$\begin{aligned} \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} &= \begin{pmatrix} 1 \\ 3 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= \begin{pmatrix} 1 \\ 4 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned}$$

Shear factor
= $\frac{\text{Dist bwn point \& image}}{\text{Dist of point from y axis}}$



Shear factor
= $\frac{AA'}{A \text{ to } y\text{-axis}}$
= $\frac{3}{1}$
= 3

$$\begin{aligned} \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= \begin{pmatrix} 1 \\ 4 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} &= \begin{pmatrix} 3 \\ 10 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} &= \begin{pmatrix} 3 \\ 11 \end{pmatrix} \end{aligned}$$

Rotation thru an angle 120° anticlockwise about origin

$$\begin{aligned}\text{Trans matrix} &= \begin{pmatrix} \cos 120^\circ & -\sin 120^\circ \\ \sin 120^\circ & \cos 120^\circ \end{pmatrix} \\ &= \begin{pmatrix} -\cos 60^\circ & -\sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{pmatrix} \\ &= \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}\end{aligned}$$

simplify as usual

If given $\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$

Step 1 (Don't come about +ve/-ve first)
Make it $\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

$$= \begin{pmatrix} -\cos 60^\circ & -\sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{pmatrix}$$

Step 2 (Deal with the +ve/-ve)
Make it $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

$$= \begin{pmatrix} \cos 120^\circ & -\sin 120^\circ \\ \sin 120^\circ & \cos 120^\circ \end{pmatrix}$$

Example 2 (clockwise)

$$\begin{pmatrix} \cos 60^\circ & \sin 60^\circ \\ -\sin 60^\circ & \cos 60^\circ \end{pmatrix}$$
$$= \begin{pmatrix} \cos(-60^\circ) & -\sin(-60^\circ) \\ \sin(-60^\circ) & \cos(-60^\circ) \end{pmatrix}$$

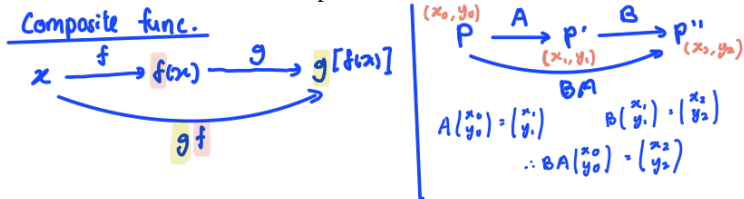
$$\begin{aligned}\cos(-\theta) &= \cos \theta \\ \sin(-\theta) &= -\sin \theta\end{aligned}$$

Clockwise rotation by angle 60°
about origin

Successive Transformations

Let matrices **A** and **B** represent the transformations *S* and *T* respectively.

Then the matrix which represents the successive transformations *S* followed by *T* is **BA**.



Inverse of a Transformation (Reversing the effect of a transformation)

Let *T* be a transformation represented by matrix **A**.

Then *T* is invertible if and only if **A** is invertible, in which case the transformation T^{-1} is represented by the matrix A^{-1} .

$(x_0, y_0) \xrightarrow{A} (x_1, y_1)$
 $A \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$

If A^{-1} exists

$A^{-1} A \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = A^{-1} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$
 $A^{-1} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$

Change of Areas after a Transformation

Let *R* be a region with area A_R . Suppose *R'* is the image of the region *R* under the transformation *T* represented by the transformation matrix **M**. Then the area of the region *R'* is given by

$$\text{Area of region } R', A_{R'} = A_R \times |\det(\mathbf{M})|$$

Region *R* $\xrightarrow{M}$ Region *R'*
 $(\text{Area } A_0)$ $(\text{Area } A_0 \times |\det(M)|)$
 Area A_0 Area A_0

Example

$\triangle ABC \xrightarrow{M} \triangle DEF$
 $(A = 10\text{cm}^2)$ (Area ?)

$$M = \begin{pmatrix} 1 & 4 \\ 7 & 3 \end{pmatrix}$$

$$\det(M) = 5 - 28 = -23$$

$$\text{Area of } \triangle DEF = 10 \times |-23| = 230\text{cm}^2$$

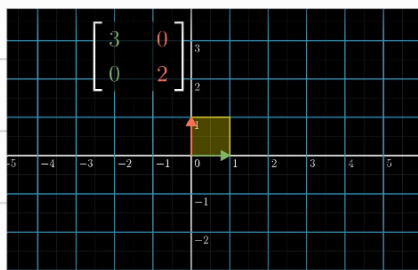
Determinant in 2D

When a ^{2D} linear transformation is performed, the area of a given region will be stretched or squished. The factor by the area of a given region increases or decreases by that transformation matrix is the determinant of that matrix.

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = (ad - bc)$$

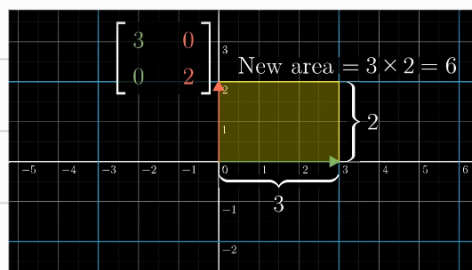
Intuition

Assume a matrix of the form $\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$ where b & c is 0



Area = 1 unit²

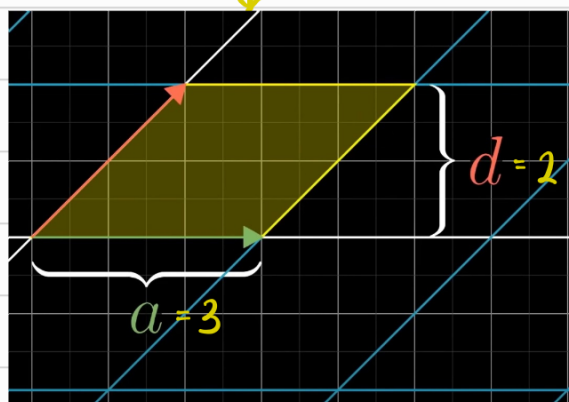
Transformation →



Area = 6 unit²

$$\therefore \text{Det} = (3)(2) - 0 = 6$$

$$\begin{aligned} \text{If } b &= 0 \\ c &= 2 \end{aligned}$$



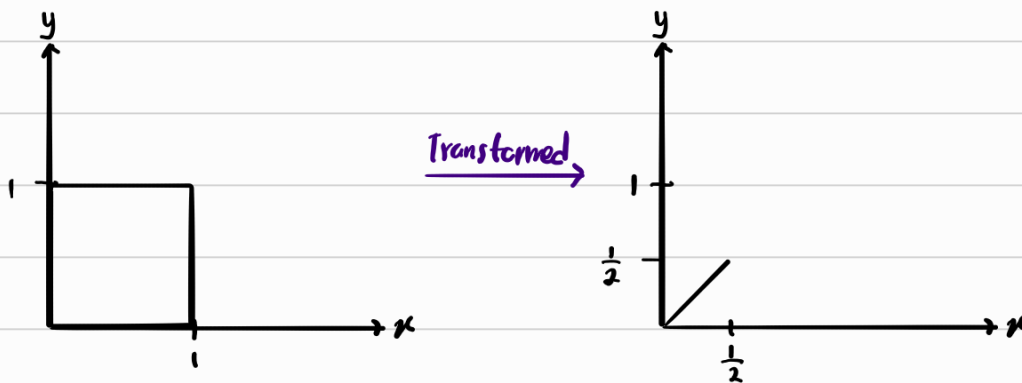
Area = 6 unit²

Example

$\hat{i}_c$ and $\hat{j}_{new}$ is linearly dependent

$$\text{Transformation matrix, } P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$\hat{i}_{new}$ and $\hat{j}_{new}$ is a parallel vec. from the origin, thus collapsing entire space to a line.



$$\text{Determinant} = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) - \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 0$$

Determinant in 3D

When a 3P linear transformation is performed, the volume of a given region will be stretched or squished. The factor by the volume of a given region increases or decreases by that transformation matrix is the determinant of that matrix.

2.3: DETERMINANT OF A MATRIX

Introduction

1. A determinant is a **scalar** which is associated with each **square matrix**. Let **A** be an $n \times n$ matrix, then the determinant of **A** is denoted by $\det(\mathbf{A})$ or $|\mathbf{A}|$.
2. **Determinant of a 2×2 matrix:**

The determinant of a 2×2 matrix $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ is given by

$$\det(\mathbf{A}) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

3. **Determinant of a 3×3 matrix:**

The determinant of a 3×3 matrix $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ is given by

$$\begin{aligned} \det(\mathbf{A}) &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\ &= +a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \end{aligned}$$

Properties of Determinants

Let $\mathbf{A}$ be a square matrix.

- (a) If two rows (or two columns) in matrix $\mathbf{A}$ are identical, then $\det(\mathbf{A}) = 0$.

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 7 \\ 1 & 2 & 3 \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} 1 & 1 & 4 \\ 2 & 2 & 5 \\ 3 & 3 & 7 \end{vmatrix} = 0$$

Row 1 & Row 3 identical Column 1 & Column 2 identical

- (b) If matrix $\mathbf{A}$ has a row of zeros (or a column of zeros), then $\det(\mathbf{A}) = 0$.

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ -1 & -4 & 5 \end{vmatrix} = 0 \quad \begin{vmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & -1 & 1 \end{vmatrix} = 0$$

- (c) (i) If a row in $\mathbf{A}$ is a multiple of another row, then $\det(\mathbf{A}) = 0$.

- (ii) If a column in $\mathbf{A}$ is a multiple of another column, then $\det(\mathbf{A}) = 0$.

$$\text{i) } \begin{vmatrix} 1 & 2 & 3 \\ 10 & 20 & 30 \\ -1 & -4 & 5 \end{vmatrix} = 0 \quad (\because R_2 = 10R_1) \quad \text{ii) } \begin{vmatrix} 10 & 1 & -1 \\ 20 & 2 & -4 \\ 30 & 3 & 5 \end{vmatrix} = 0 \quad (\because C_1 = 10C_2)$$

- (d) If $\mathbf{A}$ is an $n \times n$ **triangular matrix** (upper triangular, lower triangular, or diagonal), then $\det(\mathbf{A})$ is the **product** of the elements on the **main diagonal** of the matrix.

$$\begin{vmatrix} 1 & 2 & 4 \\ 0 & 5 & -1 \\ 0 & 0 & -2 \end{vmatrix} = (1)(5)(-2) = -10 \quad \begin{vmatrix} 1 & 0 & 0 \\ 3 & -2 & 0 \\ 5 & 0 & 4 \end{vmatrix} = 1(-2)(4) = -8 \quad \begin{vmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{vmatrix} = 5(4)(3) = 60$$

$\det = 1 \begin{vmatrix} 5 & -1 \\ 0 & -2 \end{vmatrix} = -0 + 0 = 1(5(-2) - 0) = -10$

- (e) (i) If $\mathbf{B}$ is the matrix that results when a single row or single column of $\mathbf{A}$ is multiplied by a scalar k , then $\det(\mathbf{B}) = k \det(\mathbf{A})$.

- (ii) If $\mathbf{B}$ is the matrix that results when two rows or two columns of $\mathbf{A}$ are interchanged, then $\det(\mathbf{B}) = -\det(\mathbf{A})$.

- (iii) If $\mathbf{B}$ is the matrix that results when a multiple of one row of $\mathbf{A}$ is added to another or when a multiple of one column is added to another, then

$$\det(\mathbf{B}) = \det(\mathbf{A}).$$

(f) $\det(\mathbf{A}^T) = \det(\mathbf{A})$

- (g) $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$, where $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same order.

- (h) $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$.

If $\mathbf{A}$ is invertible, then $\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}$. $\det(\mathbf{A}^{-1}) = [\det(\mathbf{A})]^{-1}$

Minor and Cofactor

1. If $\mathbf{A}$ is a square matrix, then the **minor of element** a_{ij} is denoted by M_{ij} and is defined to be the **determinant of the submatrix** that remains after the i^{th} row and j^{th} column are deleted from $\mathbf{A}$.

The number $(-1)^{i+j}M_{ij}$ is denoted by C_{ij} and is called the **cofactor of element** a_{ij} .

2. Note that a minor M_{ij} and its corresponding cofactor C_{ij} are either the same or negatives of each other and that the relating sign $(-1)^{i+j}$ is either +1 or -1 in accordance with the pattern in the “checkerboard” array

Example

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 4 \\ 6 & 7 & 9 \\ 2 & -1 & 8 \end{pmatrix} \quad \begin{matrix} M_{11} = \begin{vmatrix} 7 & 9 \\ -1 & 8 \end{vmatrix} = 56 - (-9) = 65 \\ M_{23} = \begin{vmatrix} 1 & 3 \\ 2 & -1 \end{vmatrix} = -1 - 6 = -7 \\ C_{11} = (-1)^{1+1} M_{11} = (-1)^2 (65) = 65 \\ C_{23} = (-1)^{2+3} M_{23} = (-1)^5 (-7) = -(-7) = 7 \end{matrix}$$

$$\begin{bmatrix} + & - & + & - & + & \cdots \\ - & + & - & + & - & \cdots \\ + & - & + & - & + & \cdots \\ - & + & - & + & - & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \quad \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

For example, $C_{11} = M_{11}$, $C_{21} = -M_{21}$, $C_{22} = M_{22}$

and so forth. Thus, it is never really necessary to calculate $(-1)^{i+j}$ to calculate C_{ij} — you can simply compute the minor M_{ij} and then adjust the sign in accordance with the checkerboard pattern.

Finding Determinant by Cofactor Expansion

If $\mathbf{A}$ is an $n \times n$ matrix, then the number obtained by **multiplying the elements in ANY row or column of $\mathbf{A}$** by the corresponding **cofactors** and **adding the resulting products** is called the **determinant of $\mathbf{A}$** , and the sums themselves are called cofactor expansions of $\mathbf{A}$. That is,

$$\det(\mathbf{A}) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

[cofactor expansion along the j^{th} column]

and

$$\det(\mathbf{A}) = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

[cofactor expansion along the i^{th} row]

$$\det \mathbf{A} = +1 \begin{vmatrix} 7 & 9 \\ -1 & 8 \end{vmatrix} - 3 \begin{vmatrix} 6 & 9 \\ 2 & 8 \end{vmatrix} + 4 \begin{vmatrix} 6 & 7 \\ 2 & -1 \end{vmatrix}$$

$$= +1 \begin{vmatrix} 7 & 9 \\ -1 & 8 \end{vmatrix} + (3) \left(- \begin{vmatrix} 6 & 9 \\ 2 & 8 \end{vmatrix} \right) + 4 \begin{vmatrix} 6 & 7 \\ 2 & -1 \end{vmatrix}$$

$$\det \mathbf{A} = a_{11} C_{11} + a_{21} C_{21} + a_{31} C_{31}$$

$\downarrow$
 $-M_{21}$

Example 10

Let $\mathbf{A} = \begin{pmatrix} 2 & -3 & 5 \\ 2 & 0 & 0 \\ 1 & 3 & -2 \end{pmatrix}$. Evaluate $\det(\mathbf{A})$ by cofactor expansion

- (a) along the first row of $\mathbf{A}$
 (b) along the first column of $\mathbf{A}$
 (c) along the second row of $\mathbf{A}$

Solution

$$\begin{aligned} \text{a) } \det \mathbf{A} &= 2 \begin{vmatrix} 0 & 0 \\ 3 & -2 \end{vmatrix} - (-3) \begin{vmatrix} 2 & 0 \\ 1 & -2 \end{vmatrix} + 5 \begin{vmatrix} 2 & 0 \\ 1 & 3 \end{vmatrix} \\ &= 2(0 - 0) + 3(-4 - 0) + 5(6 - 0) \\ &= 0 - 12 + 30 \\ &= 18 \end{aligned}$$

$$\begin{aligned} \text{b) } \det(\mathbf{A}) &= 1 \begin{vmatrix} 0 & 0 \\ 2 & -2 \end{vmatrix} - 2 \begin{vmatrix} -3 & 5 \\ 2 & -2 \end{vmatrix} + 1 \begin{vmatrix} -3 & 5 \\ 0 & 0 \end{vmatrix} \\ &= 2(0) - 2(6 - 15) + 1(0) \\ &= 18 \end{aligned}$$

$$\begin{aligned} \text{c) } \det(\mathbf{A}) &= -2 \begin{vmatrix} -3 & 5 \\ 3 & -2 \end{vmatrix} + 0 - 0 \\ &= -2(6 - 15) \\ &= 18 \end{aligned}$$

Finding the Inverse of a Matrix Using Its Adjoint**1. Matrix of Cofactors from A and Adjoint of A:**

If A is any $n \times n$ matrix and C_{ij} is the cofactor of a_{ij} , then the matrix

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix} \quad \begin{matrix} + & - \\ - & + \end{matrix}$$

is called the **matrix of cofactors from A**.

The **transpose** of this matrix is called the **adjoint of A** and is denoted by $\text{adj}(A)$.

2. Inverse of an $n \times n$ matrix:

If A is an invertible matrix, then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

Example 11

Given $A = \begin{pmatrix} 3 & 4 \\ 6 & -2 \end{pmatrix}$. Find $\text{adj}(A)$. Hence, find A^{-1} .

Solution

$$\text{Matrix of cofactors from } A = \begin{pmatrix} +(-2) & -(6) \\ -(4) & +(3) \end{pmatrix} = \begin{pmatrix} -2 & -6 \\ -4 & 3 \end{pmatrix}$$

$$\text{Adj}(A) = \begin{pmatrix} -2 & -4 \\ -6 & 3 \end{pmatrix}$$

$$\begin{aligned} \det(A) &= 3(-2) - (4)(6) \\ &= -6 - 24 \\ &= -30 \end{aligned}$$

$$A^{-1} = \frac{1}{-30} \begin{pmatrix} -2 & -4 \\ -6 & 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{15} & \frac{2}{15} \\ \frac{1}{5} & -\frac{1}{10} \end{pmatrix}$$

Example 12

Show that the inverse of matrix $\begin{pmatrix} 3 & 2 \\ 6 & 4 \end{pmatrix}$ does not exist.

Solution

$$\begin{vmatrix} 3 & 2 \\ 6 & 4 \end{vmatrix} = 12 - 12 = 0$$

$\therefore$ Inverse does not exist

$$\begin{aligned} A &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ M_{11} &= |d| = d \\ M_{12} &= |c| = c \\ M_{21} &= |b| = b \\ M_{22} &= |a| = a \end{aligned} \quad \begin{aligned} C_{11} &= d \\ C_{12} &= -c \\ C_{21} &= -b \\ C_{22} &= a \end{aligned} \quad \begin{aligned} &\text{Matrix of cofactor} \\ &\text{of } A \\ &= \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} \\ &= \begin{pmatrix} d & -c \\ -b & a \end{pmatrix} \\ &\text{Transpose} \\ &\text{Adj}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \end{aligned}$$

$$\text{Ans: } A^{-1} = \begin{pmatrix} \frac{1}{15} & \frac{2}{15} \\ \frac{1}{5} & -\frac{1}{10} \end{pmatrix}$$

$$2 \begin{vmatrix} 1 & 5 \\ 4 & 0 \end{vmatrix} - 0 + 6 \begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix} = \begin{pmatrix} + (0-20) - (0+5) + (12+1) \\ - (0-24) + (0+6) - (8-0) \\ + (0-6) - (10-18) + (2-0) \end{pmatrix}$$

$$2(0-20) - 0 + 6(12+1) = 38$$

Example 13

Given $A = \begin{pmatrix} 2 & 0 & 6 \\ 3 & 1 & 5 \\ -1 & 4 & 0 \end{pmatrix}$. Find $\text{adj}(A)$. Hence, find A^{-1} .

Ans: $A^{-1} = \begin{pmatrix} -\frac{10}{19} & \frac{12}{19} & -\frac{3}{19} \\ -\frac{5}{38} & \frac{3}{19} & \frac{4}{19} \\ \frac{13}{38} & -\frac{4}{19} & \frac{1}{19} \end{pmatrix}$

Solution

Minors of cofactors from A

$$= \begin{pmatrix} + (0-20) & - (0+5) & + (12+1) \\ - (0-24) & + (0+6) & - (8-0) \\ + (0-6) & - (10-18) & + (2-0) \end{pmatrix}$$

$$= \begin{pmatrix} -20 & -5 & 13 \\ 24 & 6 & -8 \\ -6 & 8 & 2 \end{pmatrix}$$

$$\text{Adj } A = \begin{pmatrix} -20 & 24 & -6 \\ -5 & 6 & 8 \\ 13 & -8 & 2 \end{pmatrix}$$

$\det(A)$

$$= +2 \begin{vmatrix} 1 & 5 \\ 4 & 0 \end{vmatrix} - 0 + 6 \begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix}$$

$$= 2(0-20) + 6(12+1)$$

$$= 38$$

$$A^{-1} = \frac{1}{38} \begin{pmatrix} -20 & 24 & -6 \\ -5 & 6 & 8 \\ 13 & -8 & 2 \end{pmatrix}$$

perform ERO to reduce to RREF

Supplement: Finding the Inverse of a Matrix Using ERO

1. **Theorem:**

An $n \times n$ matrix A is **invertible** if and only if A is **row equivalent** to I_n , and in this case, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .

2. **Algorithm for finding A^{-1} :**

Row reduce the augmented matrix $(A \mid I_n)$.

- If A is row equivalent to I_n , then $(A \mid I_n)$ is row equivalent to $(I_n \mid A^{-1})$.
- If A is *not* row equivalent to I_n , then A does not have an inverse.

Example 9

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = I \implies \begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}$$

Using elementary row operations, find the inverse of the matrix $A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{pmatrix}$, if it exists.

Solution

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array} \right]$$

$A \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ I

$$\xrightarrow{R_3 \rightarrow R_3 - 4R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{array} \right]$$

$$\xrightarrow{R_3 \rightarrow R_3 + 3R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{array} \right]$$

$$\xrightarrow{R_3 \rightarrow \frac{1}{2}R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array} \right]$$

$$\begin{array}{l} \xrightarrow{R_1 \rightarrow R_1 - 3R_3} \\ \xrightarrow{R_2 \rightarrow R_2 - 2R_3} \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 0 & -2 & -4 & -1 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array} \right]$$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ A^{-1}

$$\therefore A^{-1} = \begin{bmatrix} -\frac{9}{2} & 7 & -\frac{3}{2} \\ -2 & -4 & -1 \\ \frac{3}{2} & -2 & \frac{1}{2} \end{bmatrix}$$

Sys. of linear eqn & matrix

$$a_{11}x + a_{12}y = b_1$$

$$a_{21}x + a_{22}y = b_2$$

↓ matrix form

$$\begin{matrix} A & \underline{x} & = & \underline{b} \\ \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} & \begin{pmatrix} x \\ y \end{pmatrix} & = & \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \end{matrix}$$

Transformation Matrix, A

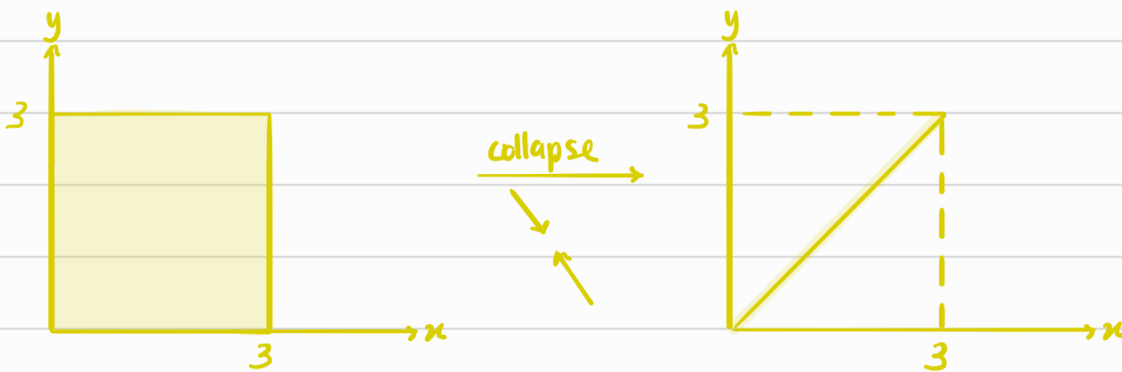
position vec.
[a point in space]

Solving this eqn means finding the point (x, y) that will be transformed to (b_1, b_2) under the transformation matrix

If matrix A is

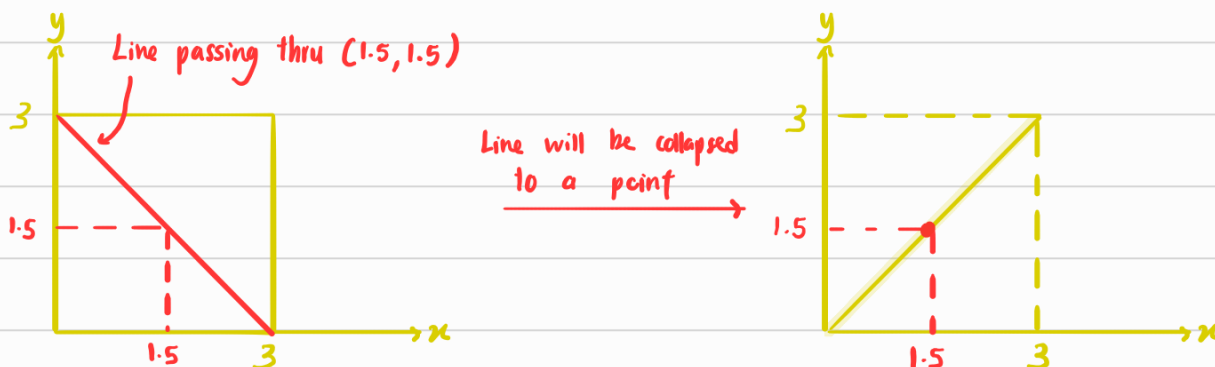
- invertible $[\det A \neq 0] \implies$ unique solution
- Not invertible $[\det A = 0] \implies$ Infinite solutions / No solutions

When space collapses into a lower dimension [eg: Plane $\rightarrow$ Line],



① there will be infinite sol. if $\underline{b}$ was a point on the collapsed line.

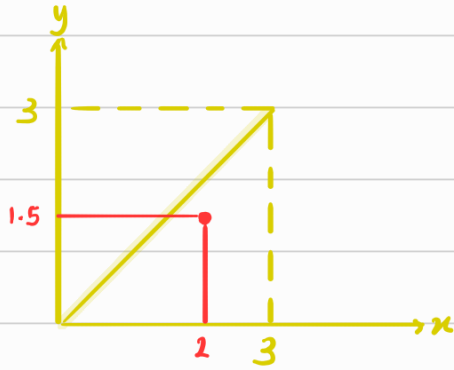
Example if $\underline{b}$ was $\begin{pmatrix} 1.5 \\ 1.5 \end{pmatrix}$



$\therefore$ Thus all the points on that line will be the solution. Since a line consists of infinitely many points, thus there are infinitely many solutions

② There will be no solutions if $\underline{b}$ was NOT a point on the collapsed line.

Example if $\underline{b}$ was $\begin{pmatrix} 2 \\ 1.5 \end{pmatrix}$



Since the entirety of space have already been collapsed into a line (Every point in space have "moved" into that line), there will be NO points left that is "moved" outside of that line.

$\therefore$ So there will be no $\underline{x}$ that becomes $\underline{b}$ under this particular transformation matrix